

$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cap B) = P(A)P(B|A)$$

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots + P(B|A_n)P(A_n)$$

$$P(A|B) = P(A)P(B|A) / P(B)$$

$$H(X) = - \sum_{i=1}^n p(x_i) \log p(x_i)$$

$$H(X,Y) = - \sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) \log p(x_i, y_j)$$

$$H(Y|X) = \sum_{i=1}^n p(x_i) H(Y|x_i)$$

$$H(Y|x_i) = \sum_{j=1}^m p(y_j|x_i) \log p(y_j|x_i)$$

$$I(X,Y) = \sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) \log \frac{p(x_i, y_j)}{p(x_i)p(y_j)}$$

$$w = \{w_1, w_2, \dots, w_K\}, k = 1, 2, \dots, K$$

$$x = (x_1, x_2, \dots, x_n)^T$$

$$L(x, w) = \prod_{i=1}^n p(x_i, w)$$

$$l(x, w) = \ln (L(x, w)) = \ln \left(\sum_{i=1}^n p(x_i, w) \right) = \sum_{i=1}^n \ln p(x_i, w)$$

$$w^* = \arg \max_w l(x, w)$$

$$\frac{\partial l}{\partial w} = 0$$

$$d(A,B) = d^2(\bar{A}, \bar{B})$$

$$\overline{AB} = \frac{n_A \bar{A} + n_B \bar{B}}{n_A + n_B}$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$ERR = \frac{FP + FN}{TP + TN + FP + FN}$$

$$TPR = \frac{TP}{TP + FN}$$

$$TNR = \frac{TN}{TN + FP}$$

$$PPV = \frac{TP}{TP + FP}$$

$$F1 = \frac{2TP}{2TP + FP + FN}$$

$$Balanced\ Accuracy = \frac{1}{2} TPR + \frac{1}{2} TNR$$